

03 Amazon MWAA Setup Guide

Amazon MWAA (Managed Workflows for Apache Airflow) Setup Guide

Prerequisites

- AWS Account with appropriate permissions
- Basic understanding of S3 and VPC concepts
- 🕒 **Time Required:** ~30 minutes for environment creation

Pre-Setup Requirements

1. S3 Bucket Setup

Before creating the MWAA environment, you need an S3 bucket with specific folders:

1. Create S3 Bucket:

- Navigate to S3 console
- Create bucket (e.g., `airflow-dags-[your-name]`)
- Choose appropriate region

2. Create Required Folders:

- `dags/` - For DAG files (can be empty initially)
- `requirements/` - For Python dependencies

3. Upload Requirements File:

- Create `requirements.txt` with content:

```
pandas
```

- Upload to `requirements/` folder
- 📝 **Note:** This installs pandas at runtime

Step-by-Step MWAA Setup

2. Environment Creation

1. Navigate to **Amazon Managed Workflows for Apache Airflow**
2. Click **"Create Environment"**

3. Environment Configuration

1. **Environment Name:** `workflows` (or your preferred name)
2. **Airflow Version:** Leave latest selected (e.g., 2.9.2)
3. **S3 Bucket Configuration:**
 - **S3 Bucket:** Enter your bucket name
 - **DAGs Folder:** `s3://your-bucket-name/dags`
 - **Requirements File:** `s3://your-bucket-name/requirements/requirements.txt`

- Leave other S3 options as default

4. Network Configuration

1. Click **"Next"** to proceed to networking
2. **VPC Setup:**
 - Click **"Create MWAA VPC"** (opens new tab)
 - **Stack Name:** `mwaa-vpc-v1` (or unique name)
 - **Parameters:** Leave all default values unchanged
 - **Permissions:** Leave as optional
 - Click **"Create Stack"**
 - ⌚ **Wait:** 2-3 minutes for CloudFormation stack creation
3. **Select VPC:**
 - Return to MWAA setup tab
 - Click refresh icon
 - Select the newly created VPC from dropdown

5. Access Configuration

1. **Web Server Access:** Select **"Public Network"**
 - Required for accessing Airflow UI via internet
2. **Security Group:** Select **"Create new security group"**

6. Environment Class Selection

1. **Environment Class:** Select **"Small"** (sufficient for course workloads)
 - **Specifications:**
 - 1 vCPU for Scheduler
 - 1 vCPU for Worker
 - 1 vCPU for Web Server
 - Supports up to 50 DAGs

7. Worker Configuration

1. **Maximum Worker Count:** Change to **3**
 - This is the scaling limit, not default count
 - Default minimum is 1 worker
 - Scales up automatically based on demand
2. **Minimum Worker Count:** Leave as **1**

8. Execution Role

1. **Execution Role:** Select **"Create a new role"**
2. **Role Name:** Change to meaningful name (e.g., `Airflow-Udemy-Role`)
3. Click **"Next"**

9. Review and Create

1. Review all configurations
2. Click **"Create Environment"**
3. ⌚ **Wait:** ~30 minutes for environment creation

Important Configuration Details

S3 Bucket Structure

```

your-airflow-bucket/
├── dags/
│   └── (DAG files will go here)
├── requirements/
│   └── requirements.txt
└── (other optional folders)
  
```

Environment Specifications

Component	Small Environment
Scheduler	1 vCPU
Worker	1 vCPU (min) to 3 vCPU (max)
Web Server	1 vCPU
Max DAGs	50

Access Details

- **Web Server:** Public network access
- **Security:** Managed security group
- **Scaling:** Auto-scaling based on workload

Post-Setup Steps

1. Verify Environment

1. Check environment status in MWAA console
2. Status should show "Available" when ready
3. Note the Airflow UI URL

2. Access Airflow UI

1. Click on environment name
2. Click **"Open Airflow UI"**
3. Authenticate using AWS credentials

3. Test Basic Functionality

1. Check if example DAGs are visible

2. Verify connections work
3. Test a simple DAG execution

Common Configuration Options

Requirements.txt Example

```
pandas==1.5.3
requests==2.28.2
boto3==1.26.137
psycopg2-binary==2.9.6
```

Environment Variables (Optional)

- Can be set in MWAA configuration
- Used for sensitive data and configuration
- Example: Database connection strings

Troubleshooting

Common Issues

1. **Environment Creation Failed:**
 - Check S3 bucket permissions
 - Verify requirements.txt format
 - Ensure VPC was created successfully
2. **DAGs Not Loading:**
 - Check S3 bucket path configuration
 - Verify DAG syntax
 - Check CloudWatch logs
3. **Access Issues:**
 - Verify public network access setting
 - Check security group configuration
 - Confirm IAM permissions

Monitoring

- **CloudWatch Logs:** Monitor environment health
- **Metrics:** Track DAG execution and performance
- **Airflow UI:** View DAG status and logs

Cost Considerations

- **Base Cost:** ~\$0.49/hour for small environment
- **Additional Costs:**
 - Worker scaling
 - Data transfer
 - CloudWatch logs
- **Optimization:** Use appropriate environment size

Security Best Practices

1. **IAM Roles:** Use least privilege principle
2. **VPC:** Consider private subnets for production
3. **Encryption:** Enable S3 and environment encryption
4. **Access Control:** Limit web server access

Next Steps

- Upload DAG files to S3 dags folder
- Configure connections in Airflow UI
- Set up monitoring and alerting
- Create your first data pipeline

Resources

- [Amazon MWAA Documentation](#)
- [Apache Airflow Documentation](#)
- [AWS MWAA Best Practices](#)

Last Updated: [Current Date] Course: [Your Course Name] ⚠ Remember: Environment creation takes ~30 minutes